

**Date:** 2026-04-30  
**Request ID:** 100048492

| Microwave     |                                 |             |         |         |
|---------------|---------------------------------|-------------|---------|---------|
| Element(s)    | Co, Pt                          |             |         |         |
| SOP#          |                                 |             |         |         |
| Acid Cocktail | 9 mL HCl, 3 mL HNO3, 6 mL H2SO4 |             |         |         |
| Rack          | 7                               |             |         |         |
| Vessel        | PTFE                            |             |         |         |
| Stir          | 90%                             |             |         |         |
| Time (min)    | Power (W)                       | T1 (°C)     | T2 (°C) | P (bar) |
| 15            | 1200                            | 180         | 65      | 110     |
| 15            | 1500                            | 240         | 65      | 110     |
| 30            | 1500                            | 240         | 65      | 110     |
| sample(s)     | weight (g)                      | volume (mL) |         |         |
| 200127788_1   | 0.5049                          | 100         |         |         |
| 200127788_2   | 0.5072                          | 100         |         |         |
| 200127789_1   | 0.5033                          | 100         |         |         |
| 200127789_2   | 0.5054                          | 100         |         |         |
| 200127790_1   | 0.5044                          | 100         |         |         |
| 200127790_2   | 0.5070                          | 100         |         |         |
| 200127791_1   | 0.5060                          | 100         |         |         |
| 200127791_2   | 0.5076                          | 100         |         |         |

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LOI temp: 900 °C

| sample    | crucible (g) | crucible + sample (g) | after 900 °C | LOI (%) | correction |
|-----------|--------------|-----------------------|--------------|---------|------------|
| 200127788 | 27.0107      | 29.0135               | 28.9257      | 4.38    | 0.9562     |

Co ICP analysis

| sample      | Dilution | conc. [mg/L] | Co_ppm [dried] | %Co [dried] |
|-------------|----------|--------------|----------------|-------------|
| 200127788_1 | 100/0.25 | 2.206        | 182780.11      | 18.28       |
| 200127788_2 | 100/0.25 | 2.297        | 189456.95      | 18.95       |

co result: 186118.53 ppm

co result: 18.61 %

co oxide factor:

co oxide result: 0.00 %

co lot: *Manufacturer: Assurance, 28-73COY exp: 2026-10-30*

Pt ICP analysis

| sample      | Dilution | conc. [mg/L] | Pt_ppm [dried] | %Pt [dried] |
|-------------|----------|--------------|----------------|-------------|
| 200127788_1 | 10/9     | 1.503        | 345.92         | 0.03        |
| 200127788_2 | 10/9     | 1.468        | 336.34         | 0.03        |

pt result: 341.13 ppm

pt result: 0.03 %

pt oxide factor:

pt oxide result: 0.00 %

pt lot: *Manufacturer: Assurance, 28-2PTY exp: 2027-02-28*

Note(s):

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LOI temp: 900 °C

| sample    | crucible (g) | crucible + sample (g) | after 900 °C | LOI (%) | correction |
|-----------|--------------|-----------------------|--------------|---------|------------|
| 200127789 | 23.3773      | 25.3792               | 25.2451      | 6.70    | 0.9330     |

Co ICP analysis

| sample      | Dilution | conc. [mg/L] | Co_ppm [dried] | %Co [dried] |
|-------------|----------|--------------|----------------|-------------|
| 200127789_1 | 100/0.23 | 2.282        | 211287.05      | 21.13       |
| 200127789_2 | 100/0.23 | 2.294        | 211515.57      | 21.15       |

co result: 211401.31 ppm

co result: 21.14 %

co oxide factor:

co oxide result: 0.00 %

co lot: *Manufacturer: Assurance, 28-73COY exp: 2026-10-30*

Pt ICP analysis

| sample      | Dilution | conc. [mg/L] | Pt_ppm [dried] | %Pt [dried] |
|-------------|----------|--------------|----------------|-------------|
| 200127789_1 | 10/9     | 1.408        | 333.15         | 0.03        |
| 200127789_2 | 10/9     | 1.448        | 341.20         | 0.03        |

pt result: 337.17 ppm

pt result: 0.03 %

pt oxide factor:

pt oxide result: 0.00 %

pt lot: *Manufacturer: Assurance, 28-2PTY exp: 2027-02-28*

Note(s):

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LOI temp: 900 °C

| sample    | crucible (g) | crucible + sample (g) | after 900 °C | LOI (%) | correction |
|-----------|--------------|-----------------------|--------------|---------|------------|
| 200127790 | 27.1488      | 29.1542               | 29.0779      | 3.80    | 0.9620     |

Co ICP analysis

| sample      | Dilution | conc. [mg/L] | Co_ppm [dried] | %Co [dried] |
|-------------|----------|--------------|----------------|-------------|
| 200127790_1 | 100/0.23 | 2.289        | 205111.09      | 20.51       |
| 200127790_2 | 100/0.23 | 2.365        | 210834.47      | 21.08       |

co result: 207972.78 ppm

co result: 20.80 %

co oxide factor:

co oxide result: 0.00 %

co lot: *Manufacturer: Assurance, 28-73COY exp: 2026-10-30*

Pt ICP analysis

| sample      | Dilution | conc. [mg/L] | Pt_ppm [dried] | %Pt [dried] |
|-------------|----------|--------------|----------------|-------------|
| 200127790_1 | 10/9     | 2.173        | 497.61         | 0.05        |
| 200127790_2 | 10/9     | 2.180        | 496.65         | 0.05        |

pt result: 497.13 ppm

pt result: 0.05 %

pt oxide factor:

pt oxide result: 0.00 %

pt lot: *Manufacturer: Assurance, 28-2PTY exp: 2027-02-28*

Note(s):

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LOI temp: 900 °C

| sample    | crucible (g) | crucible + sample (g) | after 900 °C | LOI (%) | correction |
|-----------|--------------|-----------------------|--------------|---------|------------|
| 200127791 | 26.4967      | 28.5045               | 28.421       | 4.16    | 0.9584     |

Co ICP analysis

| sample      | Dilution | conc. [mg/L] | Co_ppm [dried] | %Co [dried] |
|-------------|----------|--------------|----------------|-------------|
| 200127791_1 | 100/0.2  | 1.518        | 156508.86      | 15.65       |
| 200127791_2 | 100/0.2  | 1.640        | 168554.33      | 16.86       |

co result: 162531.59 ppm

co result: 16.25 %

co oxide factor:

co oxide result: 0.00 %

co lot: *Manufacturer: Assurance, 28-73COY exp: 2026-10-30*

Pt ICP analysis

| sample      | Dilution | conc. [mg/L] | Pt_ppm [dried] | %Pt [dried] |
|-------------|----------|--------------|----------------|-------------|
| 200127791_1 | 10/9     | 1.243        | 284.79         | 0.03        |
| 200127791_2 | 10/9     | 1.255        | 286.63         | 0.03        |

pt result: 285.71 ppm

pt result: 0.03 %

pt oxide factor:

pt oxide result: 0.00 %

pt lot: *Manufacturer: Assurance, 28-2PTY exp: 2027-02-28*

Note(s):

***A = crucible***

***B = crucible + sample***

***C = crucible + sample after drying***

$$\%LOI = ([B-A] - [C-A]) / (B-A) * 100\%$$

$$ppm\ M+ = [conc.][volume][dilution] / [weight]$$

$$\%M+ = [conc.][volume][dilution] / ([weight] * 10,000)$$

$$\%MO = [oxide\ factor] * \%M+$$

$$ppm\ M+ \ [dried] = ppm\ M+ / ([1 - (\%LOI) / 100])$$

$$\%M+ \ [dried] = \%M+ / ([1 - (\%LOI) / 100])$$

$$\%MO \ [dried] = \%MO / ([1 - (\%LOI) / 100])$$

$$\%Cr(VI) = (1.733[mL\ FAS][N\ FAS]) / ([weight])$$

$$\%Cr_2O_3 = 1.462 * [\%Cr(VI)]$$

$$\%Cr(III) = total\_Cr - Cr(VI)$$